

Improving PD-1 blockade plus chemotherapy for complete remission of lung cancer by nanoPDLIM2

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Abstract

Background

Immune checkpoint inhibitors (ICIs) and their combination with other therapies such as chemotherapy, fail in most cancer patients. We previously identified the PDZ-LIM domain-containing protein 2 (PDLIM2) as a *bona fide* tumor suppressor that is repressed in lung cancer to drive cancer and its chemo and immunotherapy resistance, suggesting a new target for lung cancer therapy improvement.

Methods

Human clinical samples and data were used to investigate *PDLIM2* genetic and epigenetic changes in lung cancer. Using an endogenous mouse lung cancer model faithfully recapitulating refractory human lung cancer and a clinically feasible nano-delivery system, we investigated the therapeutic efficacy, action mechanism, and safety of systemically administrated PDLIM2 expression plasmids encapsulated in nanoparticles (nanoPDLIM2) and its combination with PD-1 antibody and chemotherapeutic drugs.

Results

PDLIM2 repression in human lung cancer involves both genetic deletion and epigenetic alteration. NanoPDLIM2 showed low toxicity, high tumor specificity, antitumor activity, and greatly improved the efficacy of anti-PD-1 and chemotherapeutic drugs, with complete tumor remission in most mice and substantial tumor reduction in the remaining mice by their triple combination. Mechanistically, nanoPDLIM2 increased major histocompatibility complex class I (MHC-I) expression, suppressed multi-drug resistance 1 (MDR1) induction and survival genes and other tumor-related genes expression in tumor cells, and enhanced lymphocyte

tumor infiltration, turning the cold tumors hot and sensitive to ICIs and rendering them vulnerable to chemotherapeutic drugs and activated tumor-infiltrating lymphocytes (TILs) including those unleashed by ICIs.

Conclusions

These studies established a clinically applicable PDLIM2-based combination therapy with great efficacy for lung cancer and possibly other cold cancers.

eLife assessment

This study presents a **valuable** finding for the immunotherapy of cancer. The data support the role of PDLIM2 as a tumor suppressor, and more immediately, its relevance for strategies to improve the efficacy of immunotherapy. The evidence supporting the conclusions is **compelling** and the work will be of interest to biomedical scientists working on cancer immunology.

Introduction

Lung cancer is the leading cause of cancer-related deaths in both men and women with a 5-year survival rate of only 22% [1]. Although these outcomes may be improved by immune checkpoint blockade therapy involving the disruption of the binding of programmed cell death 1 (PD-1, also known as CD279) on tumor-infiltrating lymphocytes (TILs) to programmed death receptor ligand 1 (PD-L1, also known as B7-H1 or CD274) on tumor and tumor-associated cells, most lung cancer patients still fail the therapy, with a response rate of only about 20% [2]. In general, this revolutionary immunotherapy works better against ‘hot’ tumors, which have abundant TILs, strong immunogenicity and sufficient PD-L1 expression. Unfortunately, most tumors are ‘cold’, with low T-cell infiltration, weak immunogenicity and minimal PD-L1 expression, and show weak response to immune checkpoint inhibitors (ICIs) [3, 4].

Cold lung tumors without targetable oncogenic drivers are treated with chemotherapy as the standard approach [5]. However, the response rate to this conventional cancer therapy is also low and resistance often occurs after an initial response, with an overall survival (OS) of about 12–17 months. Given its roles in inducing TILs and immunogenicity, in particular PD-L1 expression, to turn cold tumors hot, chemotherapy can be an ideal candidate for combination with PD-1/PD-L1 blockade to improve therapeutic efficacy [6]. Indeed, combination treatment with ICIs and chemotherapeutic drugs shows synergy and better efficacy in both preclinical animal models and clinical trials of lung cancer and other cancers [6–8]. However, even with the combination of ICIs and chemotherapy, tumors in animals do not remit completely and the objective response rate (ORR) of lung cancer patients only reaches 33%–49.7%, with a median progression-free survival (PFS) of just 5.1–9 months and a median OS of 13–22 months [7, 8]. Thus, further improvement over the chemo-immunotherapy is direly needed.

Our recent human and mouse studies have shown that most lung tumors not only have low TILs and decreased PD-L1, but also down-regulate major histocompatibility complex class I (MHC-I), evading recognition and attack by CD8⁺ T cells, including those unleashed by ICIs and/or recruited by chemotherapy [6, 9, 10]. Following our previous cell line studies [11–17], we have established the PDZ-LIM domain-containing protein PDLIM2, also known as SLIM or mystique

[18–20], as a *bona fide* tumor suppressor and its repression as a causative driver of lung cancer and resistance to ICIs and chemotherapeutic agents [6]. While PDLIM2 is epigenetically repressed in human lung cancer, associating with therapeutic resistance and poor prognosis, its global or lung epithelial-specific deletion in mice leads to lung cancer development, chemoresistance, and complete resistance to anti-PD-1 and epigenetic drugs. One most important function of PDLIM2 is to promote the ubiquitination and proteasomal degradation of nuclear signal transducer and activator of transcription 3 (STAT3) and nuclear factor- κ B (NF- κ B) RelA (also known as p65), two master transcription factors that function as proto-oncogenes in lung and many other cancers [6, 10–13, 21–27]. PDLIM2 repression in tumor cells thus results in the persistent activation of STAT3 and RelA, leading to MHC-I downregulation and high expression of tumor growth-related genes [6]. It also leads to strong induction of multi-drug resistance 1 (MDR1) for acquired chemo-resistance, as chemotherapy further enhances RelA activation for MDR1 transcription in PDLIM2 deficient tumor cells [6].

In this study, we examined whether and how PDLIM2 can be targeted to treat lung cancer in a faithful mouse model of human lung cancer. In particular, we tested whether systemic administration of nanoparticle-encapsulated PDLIM2-expression plasmids (nanoPDLIM2) could enhance the efficacy of anti-PD-1 and/or chemotherapeutic drugs. We also examined whether PDLIM2 repression in human lung cancer involves genetic deletion and its relationship with epigenetic silencing. These studies indicate that besides epigenetic repression, loss of heterozygosity (LOH) contributes to *PDLIM2* downregulation in about 58% of human lung tumors, and that *Pdlim2* heterozygous deletion (*Pdlim2*^{+/-}) mice develop spontaneous tumors in lung and other organs. Notably, systemic administration of nanoPDLIM2 reverses the phenotypes caused by PDLIM2 repression and induces complete remission of all lung tumors in most mice without further increasing toxicity when combined with anti-PD-1 and chemotherapeutic drugs. These findings provide a firm basis to combine ICIs and chemotherapeutic drugs with PDLIM2-targeted therapy for the treatment of lung and other cancers.

Material and Methods

Animals and lung carcinogenesis

We have complied with all relevant ethical regulations for animal testing and research. The animal experiments were performed in accordance with the US National Institutes of Health (NIH) Guidelines on the Use of Laboratory Animals. All animals were maintained under pathogen-free conditions and used according to protocols approved by the Institutional Animal Care and Use Committee (IACAU) of the University of Pittsburgh. *Pdlim2*^{flx/flx}/SP-C-rtTA^{tg/-}/(tetO)7CMV-Cre^{tg/tg} (Δ SPC) mice under a pure FVB/N background and *Pdlim2*^{-/-} mice under a pure BALB/c background have been described before [6, 19, 21, 22, 28]. For lung carcinogenesis, six-to eight-week-old Δ SPC and wild type FVB/N mice were intraperitoneally (i.p.) injected with urethane (1 mg/g body weight, Sigma-Aldrich, St. Louis, MO, USA) once a week for six consecutive weeks [6], followed by different treatments as shown in the figures. Mice were euthanized at six weeks post urethane treatment for examination of lung tumors, inflammation, and treatment-induced toxicity. Surface tumors in mouse lungs were counted blinded under a dissecting microscope, and tumor diameters were measured by microcalipers.

Preparation of PDLIM2-expression plasmid or empty vector plasmid nanoparticles

The polyethylenimine (PEI)-based nanoparticles (*in vivo*-jetPEI) (Polyplus Transfection, New York, NY, USA) and plasmid DNA complexes at a nitrogen-to-phosphate ratio of 8 (N/P=8) were prepared according to the manufacturer's instructions. Briefly, 25 μ g of pCMV-myc-*Pdlim2* or empty vector plasmids in 100 μ l of a 5% glucose solution were mixed with the *in vivo*-jetPEI reagent (4 μ l)

diluted into 100Lμl of a 5% glucose solution. After 15Lminutes of incubation at room temperature, the mixed solution (200 μl/mouse) was injected intravenously (i.v.) via the tail vein. PDLIM2-expression plasmid or empty vector plasmid DNA in tissues were measured by PCR assays targeting Amp-R in the genome of these plasmids.

Histology and immunohistochemistry (IHC) analysis

Lung, liver, kidney, and spleen tissues were excised, fixed in formalin, embedded in paraffin, and cut into 4-μm-thick sections. Sections were stained with H&E or subjected to sequential incubations with the indicated primary antibodies, biotinylated secondary antibodies and streptavidin-HRP [6]. Antibodies were listed in **Supplementary Table S1**.

In vivo BrdU labeling

Mice were i.p. injected with 50Lmg/kg BrdU (Sigma-Aldrich, St. Louis, MO, USA) 24Lh prior to euthanasia. Mouse lung tissue sections were stained with anti-BrdU (Sigma-Aldrich, St. Louis, MO, USA). BrdU labeling index was calculated as the percentage of labeled cells per total cells counted (> 500 cells in each counted tumor-containing area).

Flow cytometry (FACS) analysis

The cells were isolated from mouse lungs using collagenase/dispase digestion followed by filtration with 70 μm Nylon cell strainer and red blood cell lysis; then the cells were incubated with the antibodies against cell surface antigens after blocked with αCD16/CD32. The cells were then fixed with paraformaldehyde (2%), permeabilized and incubated with antibodies against intracellular antigens if needed. For interferon-γ (IFNγ) staining, cells were treated with phorbol 12-myristate 13-acetate (PMA, 50Lng/ml), ionomycin (1LμM), brefeldin A (BFA, 3Lμg/ml) and monensin (2LμM) for 4Lh before they were stained for FACS analysis. Data were acquired and analyzed by Accuri C6 or LSRFortessa I (BD Biosciences) and FlowJo software [29].

Quantitative polymerase chain reaction (qPCR) analysis

The indicated tissues or cells were subjected to DNA or RNA extraction, RNA reverse transcription and real-time PCR using trizol, reverse transcriptase, and Power SYBR Green PCR Master Mix (Thermo Fisher Scientific, Waltham, MA USA) according to manufacturer's protocol. Primer pairs used for qPCR were listed in **Supplementary Table S2**.

Microsatellite and gene-specific PCR-based LOH analysis of *PDLIM2*

Genomic DNAs were isolated from human lung tumors and their matched normal tissues using the PureLink Genomic DNA Purification Kit (Invitrogen, Carlsbad, CA, USA), and subjected to semi-quantitative PCR using the primers specific for the microsatellite markers D8S1786 and D8S1752 that straddle the *PDLIM2* genetic locus or the *PDLIM2* genetic locus itself. Primer pairs used for the assays were listed in **Supplementary Table S2**.

Statistical analysis

One-way ANOVA power analysis was used to determine the minimum sample size. Animals were randomly assigned to different treatment groups. Measurements were taken from distinct samples. Student's t test (two tailed) and one-way ANOVA/Tukey's or two-way ANOVA/Sidak's test were used to assess significance of differences between two groups and multiple comparisons, respectively. Pearson's correlation test was used to assess association between *PDLIM2* expression with its promoter methylation or genetic deletion and the overlap between *PDLIM2* promoter methylation and genetic deletion. All bars in figures represent means ± standard error of the mean (SEM). The *p* values are indicated as **p* < 0.05, ***p* < 0.01, ns, not statistically significant, except for

those shown in figures. The p values $L < 0.05$ and 0.01 are considered statistically significant and highly statistically significant, respectively. Grubbs' and ROUT outlier tests were established a priori. There were no exclusions in the analyses.

Results

Both epigenetic repression and genetic deletion of *PDLIM2* in lung cancer

Using 40% of the expression level in matched normal lung tissues as the cut-off in the analysis of the Cancer Genome Atlas (TCGA) data, we found that *PDLIM2* was repressed in over 75% of human lung tumors in a prior study [6]. If using 50% as the cut-off, *PDLIM2* was repressed in about 93% of human lung tumors (Figure 1A). Using 125% of the methylation level of the *PDLIM2* promoter in normal lung tissues as the cut-off to analyze TCGA database, over 70% of human lung tumors were found to have hypermethylation of the *PDLIM2* promoter (Figure 1A, 1B), further validating our previous finding of epigenetic silencing as the main mechanism underlying *PDLIM2* repression [6].

In line with *PDLIM2*'s location on chromosome 8p21.3, a frequent LOH region in lung and other tumors [30–34], analysis of TCGA database revealed that *PDLIM2* expression was positively associated with its gene copy numbers, and that over 58% of human lung tumors had genetic deletion of the *PDLIM2* gene if copy number variation (CNV) of -0.1 was used as the cut-off for the gene deletion (Figure 1C). Further analysis indicated that about 44% of human lung tumors simultaneously harbored the promoter hypermethylation and LOH of the *PDLIM2* gene, and approximately 26% and 14% of them only having the promoter hypermethylation or LOH of the *PDLIM2* gene, respectively (Figure 1D). Around 16% of lung tumors possessed no such epigenetic or genetic alterations in *PDLIM2*. These findings were confirmed by microsatellite and gene-specific PCR-based LOH analysis of human primary lung tumor tissues and cell lines (Figure 1E, 1F; Supplementary Table S3). These data suggested that *PDLIM2* repression in lung cancer involves both epigenetic silencing and genetic deletion.

To determine the significance of *PDLIM2* LOH in lung tumors, we examined whether *Pdlim2* heterozygous deletion leads to decreased *PDLIM2* expression and spontaneous tumors in mice. Unlike its absolute absence in *Pdlim2* homozygous deletion (*Pdlim2*^{-/-}) mice, *PDLIM2* was detected in the lung of *Pdlim2*^{+/-} mice, but at a much lower level compared to wild type (WT) mice (Supplementary Figure S1A). Importantly, *Pdlim2*^{+/-} mice, like *Pdlim2*^{-/-} mice, also developed spontaneous tumors (Figure 1G). Of note, over 50% of tumors developed in *Pdlim2*^{+/-} and *Pdlim2*^{-/-} mice were lung tumors (Figure 1H; Supplementary Figure S1B). These data are also highly consistent with the fact that *PDLIM2* is ubiquitously expressed under physiological conditions, with the highest level in the lung and lung epithelial cells in particular [6, 18–20]. Thus, *PDLIM2* is a haploinsufficient tumor suppressor that is particularly important for lung tumor suppression.

Efficacy of systemic administration of nano*PDLIM2* in refractory lung cancer

Although reversal of *PDLIM2* epigenetic repression by epigenetic drugs to restore *PDLIM2* expression in cancer cells may be used to treat lung cancer [6], it is logical that this approach cannot be applied to lung tumors involving *PDLIM2* LOH, which accounts for about 58% of all lung cancer cases. To overcome this limitation and expand *PDLIM2*-targeted therapy to all lung tumors with *PDLIM2* repression regardless of the involved mechanisms, we tested the therapeutic efficacy of systemic administration of *PDLIM2*-expression plasmids encapsulated by the clinically feasible *in vivo*-jetPEI [35–39]. To this end, we employed mouse lung tumors induced by urethane, a

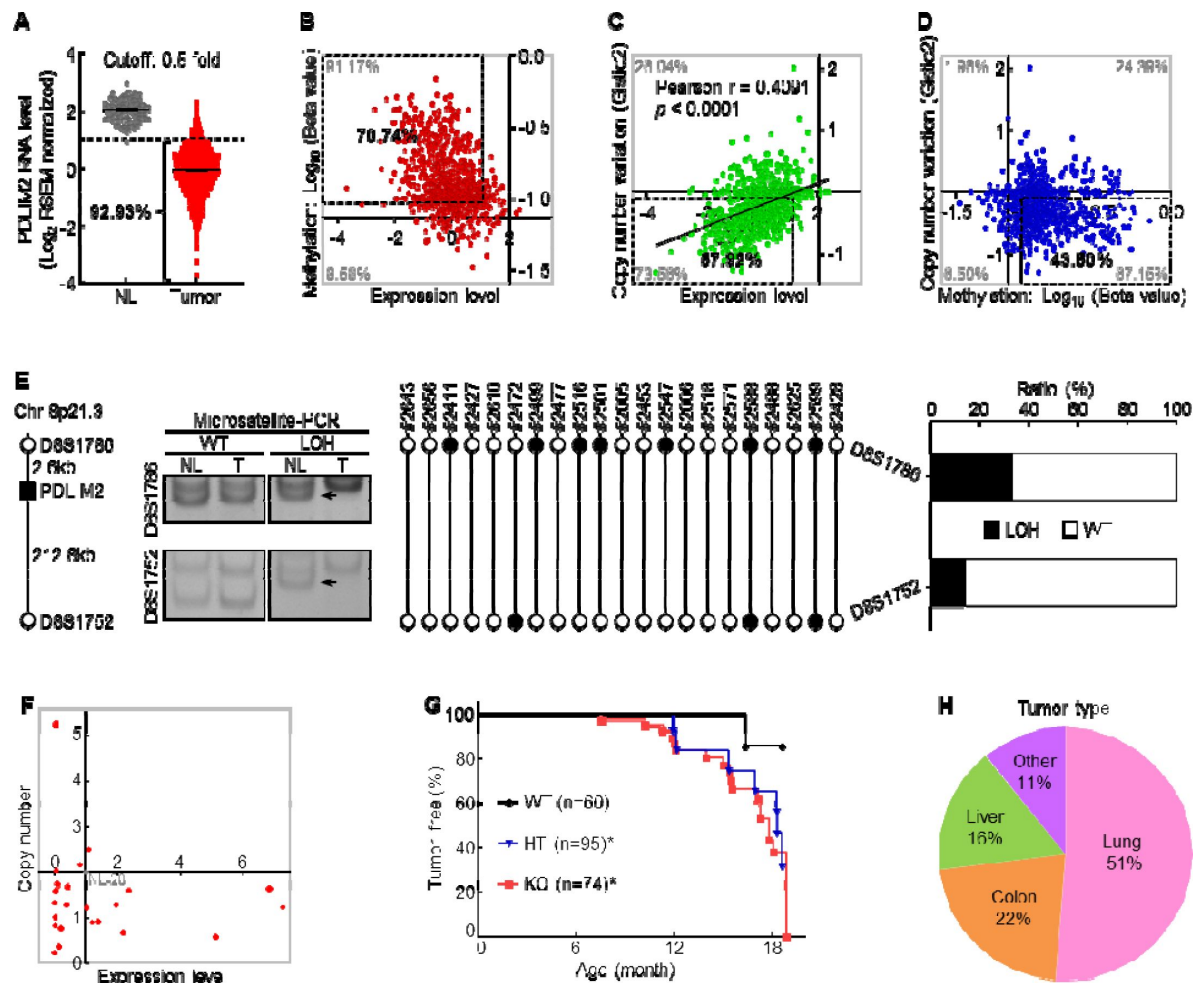


Figure 1.

PDLIM2 repression in human lung cancer involves both epigenetic alteration and genetic deletion, and *Pdlim2* genetic deletion in mice leads to development of spontaneous tumors, majorly lung tumors. (A) TCGA data showing *PDLIM2* repression in over 90% of lung tumors if using 50% of the expression level in normal lung tissues as the cut-off (NL, $n = 110$; Tumor, $n = 1019$). (B) TCGA data showing *PDLIM2* promoter hypermethylation and expression repression (dashed box) in over 70% of lung tumors when using 125% of the methylation level in normal lung tissues as the cut-off ($n = 827$). (C) TCGA data showing positive associations between *PDLIM2* expression and its gene copy numbers as well as *PDLIM2* genetic deletion and expression repression (dashed box) in about 58% of lung tumors using the copy number variation of -0.1 as the cut-off ($n = 1010$). (D) TCGA data showing simultaneous promoter hypermethylation and genomic deletion of *PDLIM2* (dashed box) in about 44% of lung tumors ($n = 816$). (E) Microsatellite-PCR showing *PDLIM2* loss in human lung tumors ($n = 21$). (F) qPCR showing *PDLIM2* loss in human lung cancer cell lines with known copy number of the *PDLIM2* gene ($n = 25$). (G) Kaplan-Meier tumor-free survival curve showing increased spontaneous tumors in *Pdlim2*^{-/-} and *Pdlim2*^{+/-} mice compared to WT mice. Gehan-Breslow-Wilcoxon test was performed. $*p < 0.05$. (H) Percentage of tumor types spontaneously developed in *Pdlim2*^{-/-} and *Pdlim2*^{+/-} mice showing a majority of lung tumors.

faithful model of human lung cancer and adenocarcinoma (AC) in particular [6, 23, 24, 40–43]. Urethane is a chemical carcinogen present in fermented food, alcoholic beverage and also cigarette smoke, the predominant risk factor accounting for about 90% of human lung cancer cases [44]. Like its human counterpart, the murine lung cancer induced by urethane also shares PDLIM2 repression, in addition to their similarities in histology, genetics, molecular biology and immunology [6, 9, 23, 24, 40–43]. WT mice with lung tumors induced by urethane were i.v. injected with nanoparticle-encapsulated *Pdlim2* plasmids or empty vector plasmids. Six weeks post the initial treatment of nanotherapy, mice were euthanized and lung tissues were collected (Figure 2A). Compared to the vector plasmid mock-treated group (Vec), nanoPDLIM2-treated mice had a significantly reduced tumor burden in their lungs, although the decrease in tumor numbers was not statistically significant (Figure 2B).

In line with the tumor reduction, nanoPDLIM2 administration decreased nuclear RelA and STAT3, a hallmark of NF- κ B and STAT3 activation, accordingly reduced the expression of their downstream cell survival gene Bcl-xL and cell proliferation gene Cyclin D1, increased apoptosis and decreased proliferation of lung cancer cells (Figure 2C–2E). These data indicated that intravenous administration of nanoPDLIM2 shows efficacy in suppressing oncogenic RelA and STAT3 activation and in treating lung cancer.

High tumor specificity and low toxicity of systemic administration of nanoPDLIM2

To characterize the potential new lung cancer therapy, we examined the expression levels of PDLIM2 in lung tumors and several organs, including the liver, kidney, and spleen. Consistent with the treatment efficacy, a high level of PDLIM2 was detected in the lung tumors from mice treated with the PDLIM2 plasmid nanoparticles one week post nanoPDLIM2 treatment, whereas no obvious PDLIM2 was found in the lung tumors from mice treated with the control plasmid nanoparticles (Figure 3A, 3B). It should be pointed out that the tumor delivery efficiency of PDLIM2-expression or control plasmid nanoparticles was similarly high (Figure 3B). However, either PDLIM2-expression or control plasmids were hardly detected in other organs/tissues of the same mice, including liver, kidney and spleen. Of note, in the first two days after injection, both empty vector and PDLIM2-expression plasmids were also detected at high levels in those tissues but quickly cleared afterward (data not shown), consistent with the well-documented tumor-specific enrichment of nanoparticles [37, 45–47]. Consistently, ectopic *Pdlim2* was not detected, and the levels of PDLIM2 proteins were comparable in these tissues (Figure 3B, 3C).

In further support of the lung tumor-specific delivery, the i.v. injection of either PDLIM2-expression or control plasmid nanoparticles showed no obvious toxicity to animals, as evidenced by no significant changes in the animal body weight and histology of all organs/tissues we examined, including spleen, kidney and liver (Figure 3D, E). Mouse appearance and behaviors, such as eating, drinking, defecating, urinating, sniffing, grooming, and digging, were not different between PDLIM2-expression and control plasmid nanoparticle groups (data not shown). Taken together, these data suggested the therapeutic efficacy and low toxicity of PDLIM2-based nanotherapy in the mouse model of refractory lung cancer.

Synergy of nanoPDLIM2 with chemotherapy in lung cancer treatment

Given the role of PDLIM2 in inhibiting the expression of cell survival and proliferation genes in tumor cells, which contribute to chemoresistance, we tested whether nanoPDLIM2 increases the efficacy of chemotherapy in the mouse model of lung cancer. Treatment with carboplatin and paclitaxel, two chemotherapeutic drugs that are often used together as the first-line treatment for lung and many other cancers, led to significant decrease in tumor number and tumor burden (Figure 4A). Importantly, combination with nanoPDLIM2 further significantly decreased both

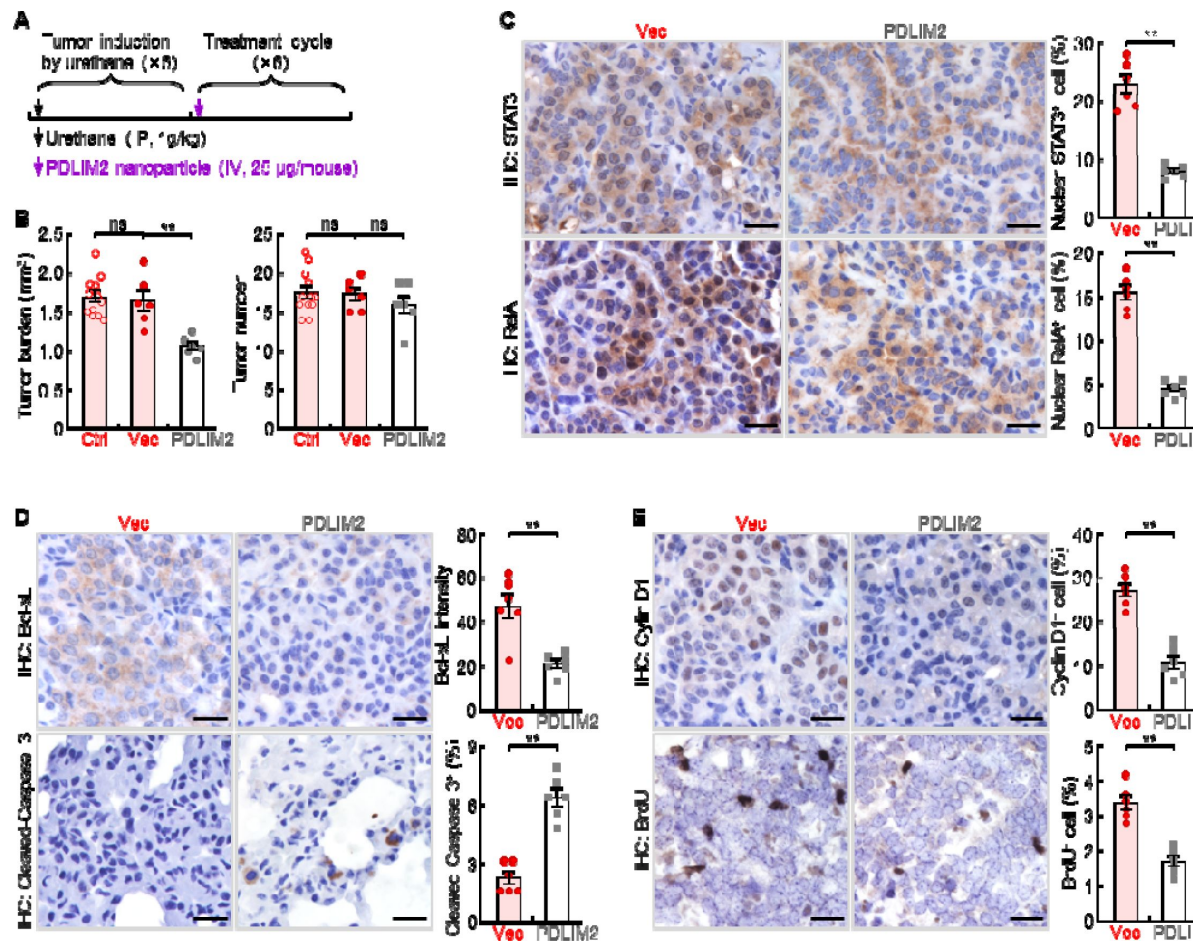


Figure 2.

Systemic administration of PDLIM2 plasmid nanoparticles shows efficacy in mouse model of refractory lung cancer. (A) Schedule of lung cancer induction and treatment. (B) Urethane model showing efficacy of intravenous administration of PDLIM2-expression plasmid nanoparticles for refractory lung cancer ($n \geq 6$). Nanoparticles with an empty vector plasmid (Vec) that was employed to express PDLIM2 were used as a control. (C) IHC staining showing decreased nuclear expression of STAT3 and RelA in lung tumors by PDLIM2 nanotherapy ($n = 6$). (D) IHC staining showing decreased Bcl-xL and increased apoptosis marker cleaved caspase-3 in lung tumors by PDLIM2 nanotherapy ($n = 6$). (E) IHC staining showing decreased Cyclin D1 and proliferation (BrdU incorporation) in lung tumors by PDLIM2 nanotherapy ($n = 6$). Scale bar in (C-E), 20 µm. Student's t test was performed (two tailed, unpaired) and data represent means $\pm$ LSEM in (B-E). $^{**}p < 0.01$; ns, not statistically significant.

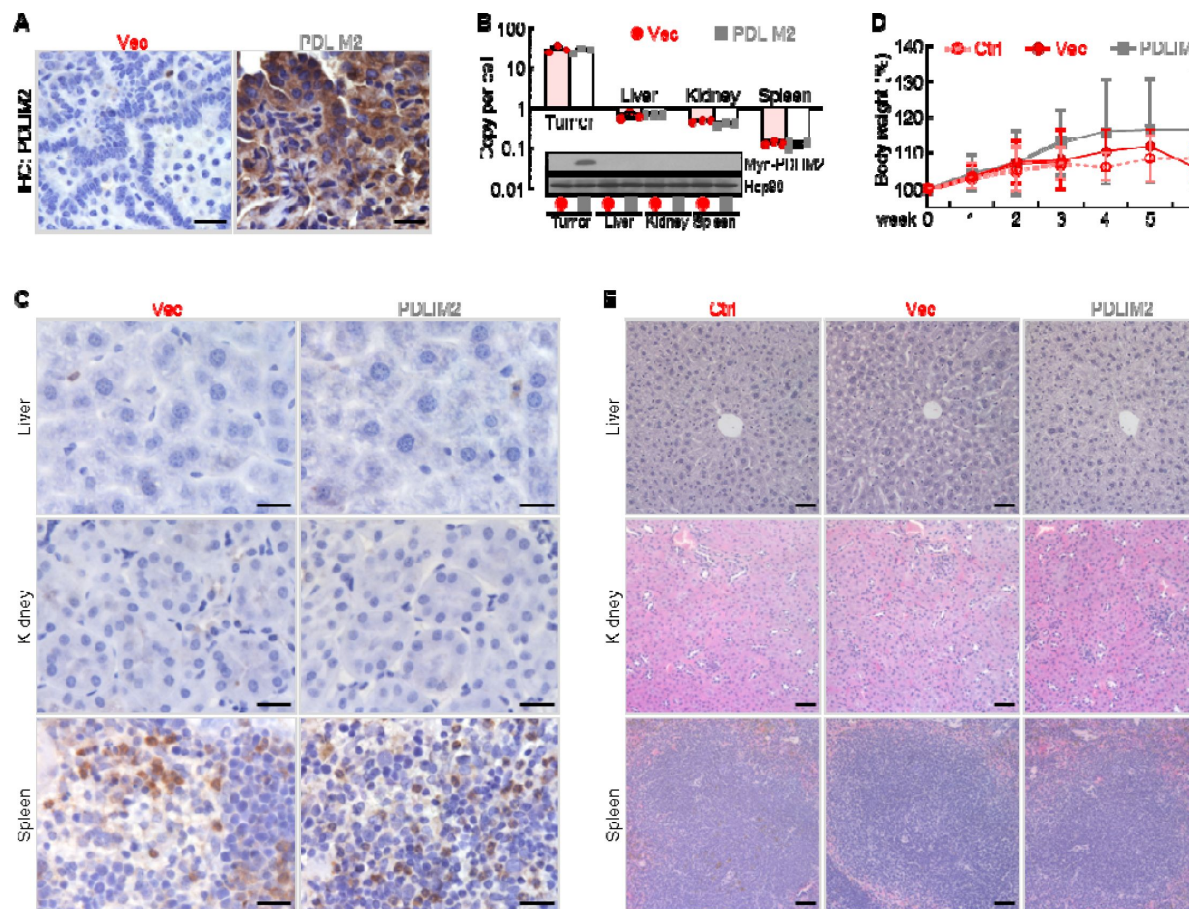


Figure 3.

PDLIM2 nanotherapy shows high tumor specificity and low toxicity. (A) IHC staining showing high PDLIM2 re-expression in lung tumors after PDLIM2 nanotherapy. (B) PCR and IB assays showing lung tumor-specific plasmids delivery and PDLIM2 expression by PDLIM2 nanotherapy (n = 3). (C) IHC staining showing comparable expression of PDLIM2 in the indicated tissues of mice treated with PDLIM2 expression plasmid or empty vector plasmid nanoparticles. (D) No significant changes in animal body weight by nanoPDLIM2 (n = 5). (E) H&E staining showing no noticeable changes in major organs by nanoPDLIM2. Scale bar: (A and C) 20 μ m; (E) 50 μ m.

tumor number and tumor burden, suggesting a promising synergy between PDLIM2 nanotherapy and chemotherapy in lung cancer treatment. Consistently, significantly higher tumor cell apoptosis was detected in mice treated with the combination therapy, in comparison to those with nanoPDLIM2 or chemotherapy alone (**Figure 4B** [↗](#)).

Another important mechanism contributing to the synergy between these two therapies involves nanoPDLIM2 blockade of the acquired chemoresistance of lung cancer cells. Chemotherapy induced strong RelA activation/nuclear expression and MDR1 expression (**Figure 4C** [↗](#), **4D** [↗](#)). NanoPDLIM2 not only repressed the constitutive activation of RelA in cancer cells but also prevented the strong induction of RelA activation and MDR1 expression by the chemotherapy. Thus, PDLIM2 nanotherapy improves the therapeutic efficacy of chemotherapy through blocking both intrinsic and acquired chemoresistance of lung cancer cells.

NanoPDLIM2 enhancement of ICI's efficacy in lung cancer treatment

High expression of cell survival genes also renders tumor cells resistant to the tumoricidal activity of cytotoxic T lymphocytes (CTLs), including those unleashed by ICIs. PDLIM2 nanotherapy should also enhance the efficacy of immunotherapy, given its ability in suppressing the expression of cell survival genes. Furthermore, NanoPDLIM2 increased the number of TILs and the expression of MHC-I, the most important and core components of immunotherapies including PD-1 immune checkpoint blockade therapy (**Figure 5A** [↗](#), **5B** [↗](#)). In line with our previous studies [[6](#) [↗](#)], PD-1 blocking antibody showed some efficacies in the mouse model of lung cancer, as evidenced by the significant decrease in tumor burden and tumor number (**Figure 5C** [↗](#)). Combining with PDLIM2 nanotherapy further significantly decreased tumor burden, although the decrease in tumor number failed to reach statistical significance. Consistently, significant increases in both CD4⁺ and CD8⁺ TILs, CD8⁺ CTL activation and lung tumor cell death were detected (**Figure 5D** [↗](#)–**5F** [↗](#)).

Complete remission of all lung tumors in most mice by the combination treatment of nanoPDLIM2 and anti-PD-1 and chemotherapeutic drugs

Like most human lung tumors, lung tumors in our animal model exhibit decreased expression of PD-L1 on cell surface, in addition to the low expression of MHC-I and low number of TILs [[6](#) [↗](#), [9](#) [↗](#)]. In line with our previous finding that PD-L1 expression is largely independent of PDLIM2 [[6](#) [↗](#)], nanoPDLIM2 failed to induce PD-L1 expression in lung tumors, which was in contrast to chemotherapy (**Figure 6A** [↗](#)). This may explain why the enhancing effect of nanoPDLIM2 on PD-1 blockade therapy is only moderate.

Although it dramatically increases TILs as well [[6](#) [↗](#)], chemotherapy also induced the expression of PD-L1 on cancer cells (**Figure 6A** [↗](#)), which presumably protects cancer cells from immune attack and thereby restricts further efficacy improvement of its combination with PDLIM2 nanotherapy. But it may increase the sensitivity of cancer cells to PD-1 blockade therapy, particularly in combination with nanoPDLIM2. Indeed, chemotherapy and PD-1 blockade therapy showed a promising synergy in reducing both tumor number and tumor burden in comparison to the individual treatment of chemotherapy or PD-1 blockade therapy [[6](#) [↗](#)]. The synergy was largely blocked when *Pdlim2* was genetically deleted from lung cancer cells (ΔSPC) (**Figure 6B** [↗](#), **6C** [↗](#)), suggesting an important role of PDLIM2 in the combination therapy of chemotherapeutic drugs and anti-PD-1. Whereas PDLIM2 nanotherapy increased MHC-I expression on tumor cells (**Figure 5B** [↗](#)), which is critical for better recognition and killing of tumor cells by CD8⁺ CTLs, chemotherapy alone or its combination with anti-PD-1 failed to do so (**Figure 6D** [↗](#)). Accordingly,

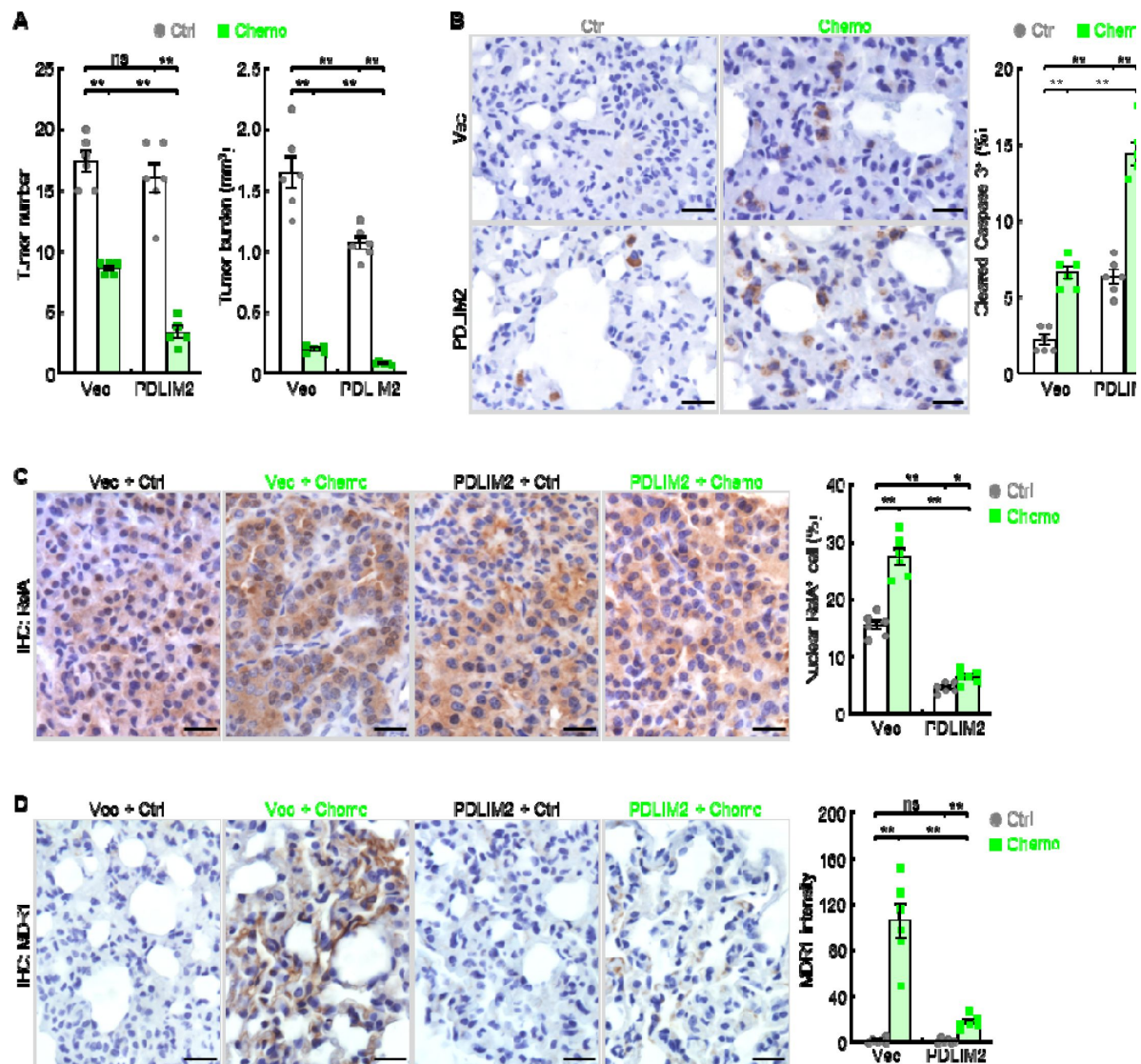


Figure 4.

PDLIM2 nanotherapy renders lung cancers more vulnerable to chemotherapy. (A) Urethane model showing synergy of PDLIM2 nanotherapy and chemotherapy in lung cancer treatment ($n \geq 5$). (B) IHC staining showing increased lung tumor cell apoptosis by PDLIM2 nanotherapy, chemotherapy, and further increase by their combination ($n = 6$). (C) IHC staining showing RelA activation by chemotherapy and blockage of chemo activation of RelA by PDLIM2 nanotherapy ($n = 6$). (D) IHC staining showing strong MDR1 induction by chemotherapy and blockage of MDR1 induction by PDLIM2 nanotherapy ($n = 6$). Scale bar in (B-D), 20 μ m. Student's t test was performed (two tailed, unpaired) and data represent means $\pm$ SEM. * $p < 0.05$; ** $p < 0.01$; ns, not statistically significant.

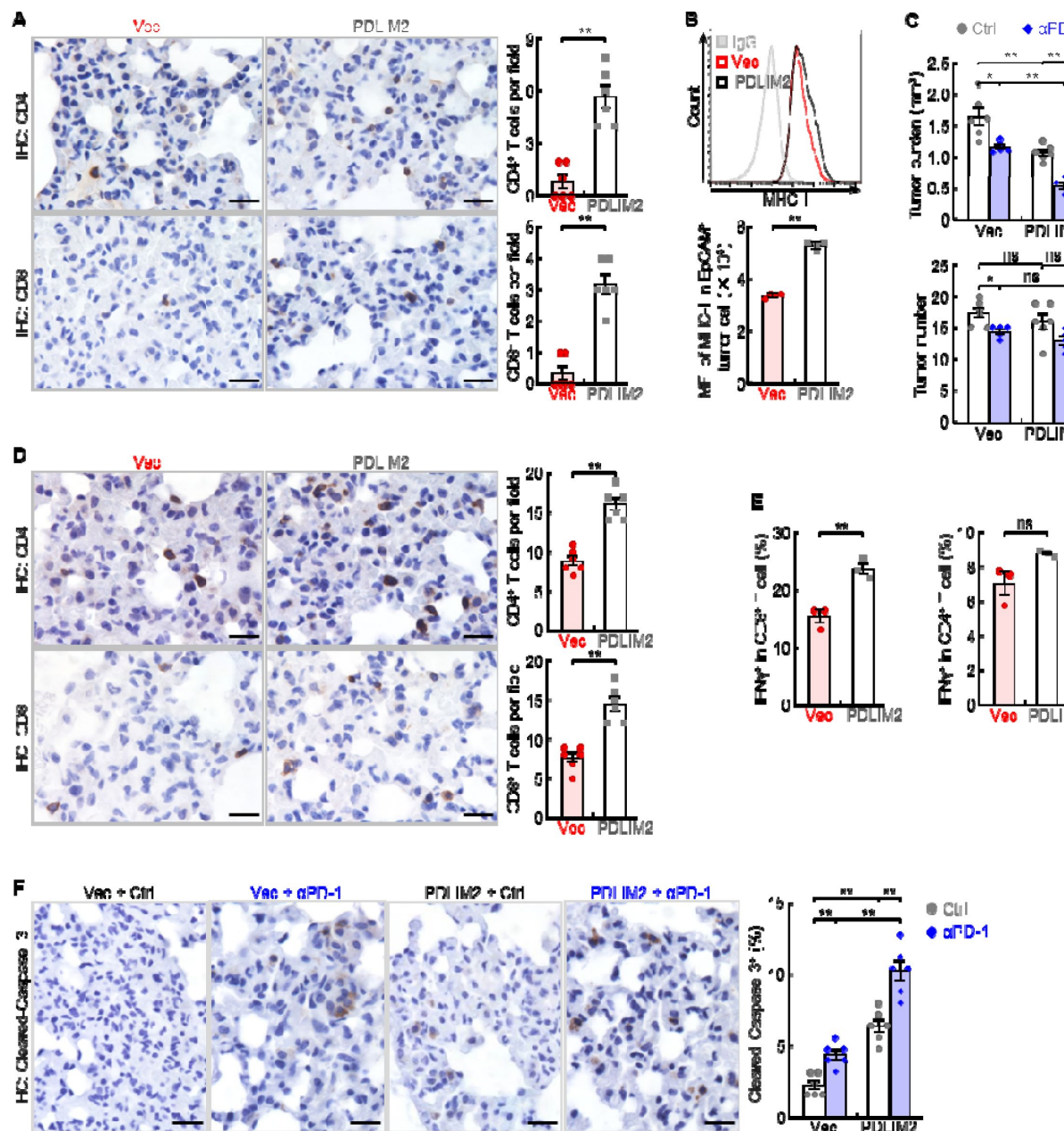


Figure 5.

PDLIM2 nanotherapy increases the efficacy of PD-1 blockade immunotherapy for refractory lung cancer. (A) IHC staining showing increased TILs in lung tumors by PDLIM2 nanotherapy (n = 6). FACS showing increased MHC-I expression in lung tumor cells by PDLIM2 nanotherapy (n = 4). Urethane model showing PDLIM2 nanotherapy enhancing PD-1 immunotherapy efficacy in lung cancer treatment (n ≥ 5). (D) IHC staining showing increased TILs by PDLIM2 nanotherapy in the context of immunotherapy (n = 6). (E) FACS showing increased activation of CD8⁺ T cells by PDLIM2 nanotherapy in the context of immunotherapy (n = 3). (F) IHC staining showing increased lung tumor cell apoptosis by PDLIM2 nanotherapy, immunotherapy, and further increase by their combination (n = 6). Scale bar in (A, D, F), 20 μm. Student's *t* test was performed (two tailed, unpaired) and data represent means ±LSEM. ***p* < 0.01; ns, not statistically significant.

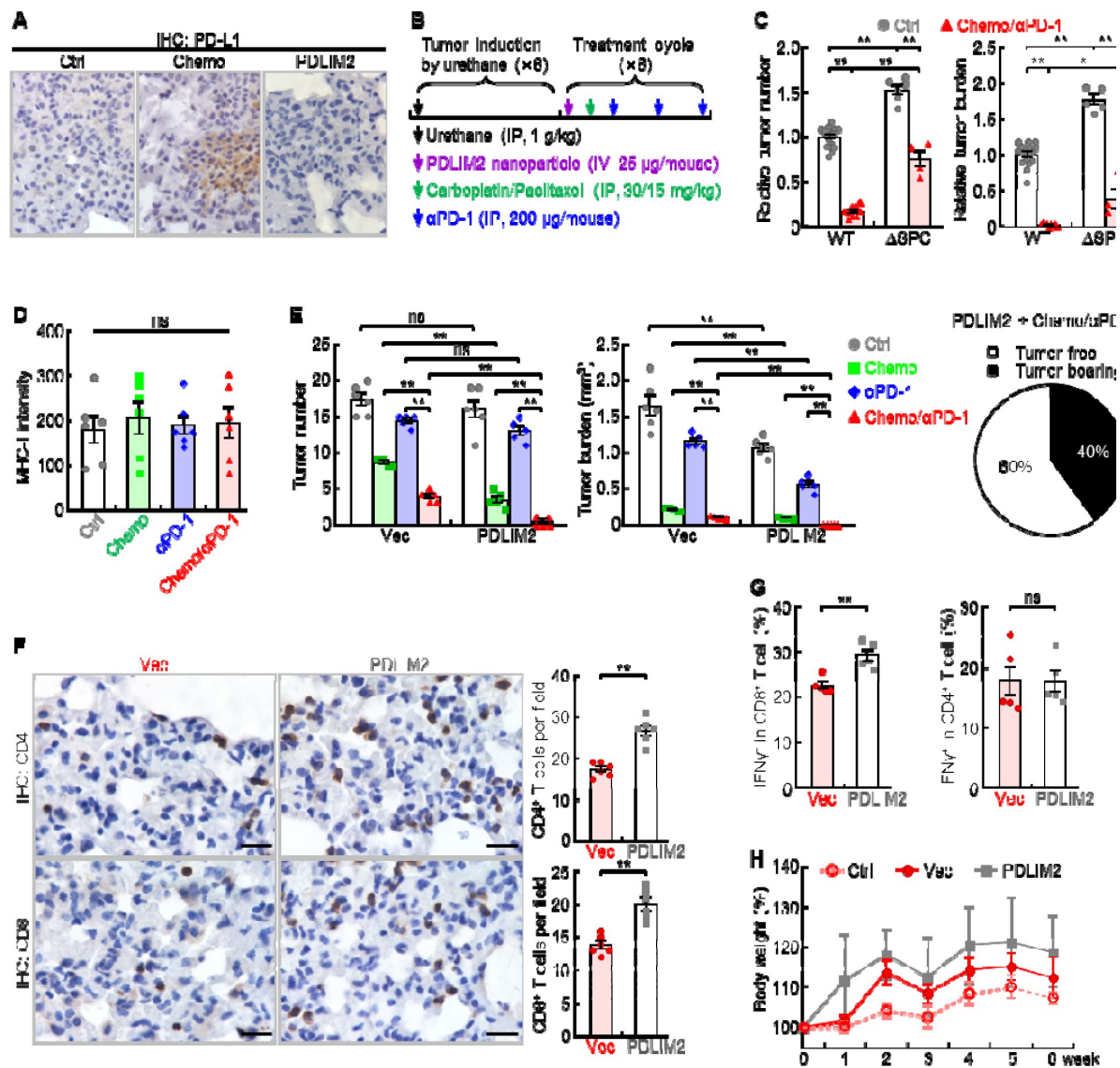


Figure 6.

Combination of PDLIM2 nanotherapy, chemotherapy and immunotherapy shows great efficacy in lung cancer treatment. (A) IHC staining showing PD-L1 induction by chemotherapy but not PDLIM2 nanotherapy. (B) Schedule of lung cancer induction and treatment. (C) Urethane model showing high resistance of lung tumors to the chemo and α PD-1 combination therapy in lung epithelial specific PDLIM2 deletion mice (Δ SPC) ($n \geq 4$). (D) IHC staining showing no MHC-I induction by chemotherapy, PD-1 immunotherapy or their combination ($n = 6$). (E) Tumor examination showing complete remission of all lung tumors in 60% of mice by combination of the three therapies ($n \geq 5$). (F) IHC staining showing increased TILs by PDLIM2 nanotherapy in mice treated with anti-PD-1 and chemotherapeutic drugs ($n = 6$). (G) FACS analysis showing increased lung CD8 $^{+}$ T-cell activation by PDLIM2 nanotherapy in mice treated with anti-PD-1 and chemotherapeutic drugs ($n = 5$). (H) No significant effect of PDLIM2 nanotherapy on the body weight of mice treated with anti-PD-1 and chemotherapeutic drugs ($n = 5$). Scale bar in (A and F), 20 μ m. Student's t test was performed (two tailed, unpaired) and data represent means $\pm$ LSEM in (c-g). * $p < 0.05$; ** $p < 0.01$; ns, not statistically significant.

combination of anti-PD-1 and chemotherapeutic drugs, like the combination therapies of nanoPDLIM2 and anti-PD-1 or chemotherapeutic drugs, also failed to induce a complete remission of lung tumors in any mice.

Given their overlapping roles in increasing TILs and in particular their complement roles in inducing MHC-I and PD-L1 expression on tumor cells, which turn tumors hot and more sensitive to PD-1 blockade, combination of nanoPDLIM2 and chemotherapeutic drugs with anti-PD-1 is expected to show better efficacy compared to combining only two of them. Indeed, combination of all three showed the strongest effect on reducing tumor burden and numbers, and caused complete remission of all lung tumors in 60% of mice (**Figure 6E**) and much more shrinkage of lung tumors in remaining mice in comparison to those treated with two combined therapies. In line with the high therapeutic efficacy, the triple combination therapy significantly increased the numbers and/or activation of CD4⁺ and CD8⁺ T cells in the lung, compared to the combination of chemotherapeutic drugs and anti-PD-1 (**Figure 6F**, **6G**).

Of note, consistent with the undetectable toxicity of PDLIM2 nanotherapy, its co-treatment did not further increase the toxicity of anti-PD-1 and chemotherapeutic drugs. There was no obvious histological difference of major organs, including the liver, lung, kidney, and spleen (**Supplementary Figure S2**). Moreover, no significant differences in animal body weights were observed by additional nanoPDLIM2, in comparison to the mice received the combinational treatment of anti-PD-1 and chemotherapeutic drugs in the presence or absence of empty vector plasmid nanoparticles (**Figure 6H**). These data suggested a novel combination therapy with very high therapeutic efficacy and no increased toxicity for lung cancer, particularly refractory lung cancer.

Discussion

PD-1/PD-L1 blockade immunotherapy has recently joined chemotherapy as a standard treatment for lung and several other cancers [2–4]. While some patients have shown dramatic responses, most patients do not benefit from this novel treatment. Currently, various combinations of ICIs with other therapies, in particular its combination with chemotherapeutic drugs are being extensively tested in both preclinical and clinical trial studies to expand the benefit of this innovative therapy [6–8]. Although a promising synergy and better efficacy has been observed in both preclinical animal models and human clinical trial studies [6–8], significant further improvement is direly needed. Using an authentic mouse model of lung cancer, we show, for the first time, that PDLIM2 nanotherapy shows efficacy and high safety, and more importantly, induces complete remission of all lung tumors in most animals when it is combined with anti-PD-1 and chemotherapeutic drugs.

Most human lung tumors as well as lung tumors in our animal model have low numbers of TILs and decreased expression of PD-L1 and MHC-I on the cell surface [6, 9], all of which are important mechanisms contributing to the resistance to PD-1 blockade therapy. Through inducing immunogenic cell death (ICD) of cancer cells [8], chemotherapy can increase TILs and PD-L1 expression on tumor cells, and thereby synergize with anti-PD-1. However, chemotherapy cannot induce MHC-I expression, which limits further improvement of its synergy with immune therapy for complete cancer remission. Also, tumor cells usually express high levels of survival genes against the tumoricidal effects of chemotherapeutic drugs and of CTLs, including those activated by chemotherapy and unleashed by ICIs.

On the other hand, PDLIM2 nanotherapy induces MHC-I expression and lymphocyte tumor infiltration but does not up-regulate PD-L1. Moreover, PDLIM2 nanotherapy prevents the induction of MDR1 and the expression of tumor-related genes and in particular cell survival genes, further sensitizing tumor cells to the cytotoxicity of chemotherapeutic drugs and immune cells

including those recruited by chemotherapy and unleashed by PD-1 blockade. Because of these important functions of PDLIM2 nanotherapy, PDLIM2 nanotherapy improved the efficacy of chemotherapy and PD-1 blockade therapy, and in combination with chemotherapy and PD-1 blockade therapy, resulted in complete cancer remission in most of the animals and dramatic tumor reduction in the remaining mice.

Another important clinical characteristic of PDLIM2 nanotherapy is its tumor-specificity and high safety profile. It delivered PDLIM2-expression plasmids to lung tumor tissues and showed undetectable toxicity in the animal model. Its combination does not further increase the toxicity of anti-PD-1 and chemotherapeutic drugs. This is in sharp contrast to the Food and Drug Administration (FDA)-approved epigenetic drugs, which can restore PDLIM2 expression in cancer cells with PDLIM2 epigenetic repression [6, 10–15]. Epigenetic drug treatment leads to body weight loss of animals (**Supplementary Figure S3A**). The toxicity is much worse when epigenetic drugs are combined with chemotherapy (**Supplementary Figure S3**).

Besides its side effects, epigenetic therapy should have much more limited application, even though it shows better efficacy as a monotherapy or combined with chemotherapeutic drugs or anti-PD-1 in comparison to PDLIM2 nanotherapy (**Supplementary Figure S4**). While nanoPDLIM2-based combination therapies could be applicable to all lung tumors with PDLIM2 repression regardless of the mechanisms involved, epigenetic therapy may be used to treat about 26% of lung tumors with PDLIM2 epigenetic repression only. About 58% of lung tumors harboring PDLIM2 LOH are not suitable to epigenetic therapies, although most of them are also with epigenetic alterations of the *pdlim2* gene. Of note, the therapeutic efficacy of epigenetic therapy depends on PDLIM2 expression [6], and PDLIM2 heterozygous loss causes spontaneous lung and other cancers. Further, the efficacy and toxicity of the combination of epigenetic agents with both anti-PD-1 and chemotherapeutic drugs has not been examined yet.

Conclusions

In summary, the presented data identify genetic deletions as a major mechanism other than epigenetic alterations for PDLIM2 repression in human lung cancer, and PDLIM2 as a haploinsufficient tumor suppressor particularly important for suppressing lung cancer and therapy resistance. More importantly, these preclinical data establish a novel combination treatment of nanoPDLIM2, anti-PD-1 and chemotherapeutic drugs that induces complete remission of all lung tumors in most animals and is also with high safety profile. We believe that these knowledges are applicable to other cancers, because PDLIM2 repression has also been linked to numerous human cancers other than lung cancer.

Abbreviations

- AC
adenocarcinoma
- BFA
brefeldin A
- CNV
copy number variation
- CTL
cytotoxic T lymphocyte
- FACS
fluorescent activated cell sorting
- FDA
Food and Drug Administration

- IFN γ
- interferon- γ
- ICD
- immunogenic cell death
- ICI
- immune checkpoint inhibitor
- IHC
- immunohistochemistry
- i.p.
- intraperitoneal
- i.v.
- intravenous
- LOH
- loss of heterozygosity
- MDR1
- multi-drug resistance 1
- MHC-I
- major histocompatibility complex class I
- nanoPDLIM2
- nanoparticle-encapsulated PDLIM2-expression plasmids
- NF- κ B
- nuclear factor- κ B
- NIH
- National Institutes of Health
- ORR
- objective response rate
- OS
- overall survival
- PD-1
- programmed cell death 1
- PD-L1
- programmed death ligand 1
- PD-L2
- programmed death ligand 2
- PDLIM2
- PDZ-LIM domain-containing protein 2
- PEI
- polyethylenimine
- PFS
- progression-free survival
- PMA
- phorbol 12-myristate 13-acetate
- qPCR
- quantitative polymerase chain reaction
- SEM
- standard error of the mean
- STAT3
- signal transducer and activator of transcription 3
- TCGA
- The Cancer Genome Atlas
- TIL
- tumor-infiltrating lymphocyte
- WT
- wild type.

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Availability of data and materials

TCGA lung adenocarcinoma, lung squamous cell carcinoma, and lung cancer data we analyzed were obtained from <https://portal.gdc.cancer.gov/projects>. All data generated or analyzed during this study are included in this article and its additional files. Extra data that support the findings of this study are available from the corresponding authors upon reasonable request.

Author contributions

Z.Q. and G.X. conceived and designed the study, led and contributed to all aspects of the analysis, and wrote the manuscript. S.D.S. contributed to the overall experimental design and data analysis and interpretation, provided advice and constructive feedback, and edited the manuscript. F.S. performed and analyzed all the experimental assays. H.Z. contributed to data analysis and edited the manuscript. P.Y. originally identified PDLIM2 LOH in human lung tumors by gene-specific PCR-based LOH analysis. Y.X. contributed to mouse clone maintenance and IHC staining.

Ethics approval and consent to participate

All animal studies were approved by the Institutional Animal Care and Use Committee of the University of Pittsburgh and carried out in accordance with NIH guidelines on animal care.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Antibody	Company	Cat. No.	Dose	Usage
Anti-PDLIM2	Everest Biotech, Ramona, CA, USA	EB11878	1:400	IHC
Anti-Cleaved Caspase 3	Cell Signaling Technology, Danvers, MA, USA	9661	1:200	IHC
Anti-Bcl-xL	Cell Signaling Technology, Danvers, MA, USA	2764	1:2400	IHC
Anti-Cyclin D1	Santa Cruz Biotechnology, Dallas, TX, USA	sc-450	1:200	IHC
Anti-BrdU	Sigma-Aldrich, St. Louis, MO, USA	B2531	1:500	IHC
Anti-RelA	Cell Signaling Technology, Danvers, MA, USA	8242	1:500	IHC
Anti-STAT3	Cell Signaling Technology, Danvers, MA, USA	4904	1:1000	IHC
Anti-MDR1	Cell Signaling Technology, Danvers, MA, USA	13978	1:800	IHC
Anti-CD4	Abcam, Cambridge, MA, USA	ab183685	1:1000	IHC
Anti-CD8	Abcam, Cambridge, MA, USA	ab209775	1:1000	IHC
Anti-PD-L1	Cell Signaling Technology, Danvers, MA, USA	64988	1:200	IHC
Anti-MHC-I	Abcam, Cambridge, MA, USA	ab15681	1:200	IHC
Anti-mouse IgG Biotinylated	Vector Laboratories, Burlingame, CA, USA	BMK-2202	1:200	IHC
Goat anti-rat IgG Biotinylated	Vector Laboratories, Burlingame, CA, USA	BA9401	1:100	IHC
Rabbit anti-goat IgG Biotinylated	Santa Cruz Biotechnology, Dallas, TX, USA	sc-2774	1:200	IHC
Goat anti-rabbit IgG Biotinylated	Dako, Carpinteria, CA, USA	E0432	1:200	IHC
Anti-Hsp90	Santa Cruz Biotechnology, Dallas, TX, USA	sc-13119	1:1000	WB
Anti-c-Myc	Santa Cruz Biotechnology, Dallas, TX, USA	sc-40	1:1000	WB
Anti-CD16/CD32	eBioscience, San Diego, CA, USA	14-0161-85	1.0 µl per sample	FACS
Anti-CD45 FITC	Biolegend, San Diego, CA, USA	103107	0.5 µl per sample	FACS
Anti-EpCAM PE	eBioscience, San Diego, CA, USA	12-5791-82	0.625 µl per sample	FACS
Anti-MHC-I APC	eBioscience, San Diego, CA, USA	17-5998-82	1.25 µl per sample	FACS
Anti-CD3 PE	eBioscience, San Diego, CA, USA	12-0031-83	2.5 µl per sample	FACS
Anti-CD4 PE-Cy7	eBioscience, San Diego, CA, USA	25-0042-82	1.25 µl per sample	FACS
Anti-CD8a APC	eBioscience, San Diego, CA, USA	17-0081-83	0.625 µl per sample	FACS
Anti-IFNγ FITC	eBioscience, San Diego, CA, USA	11-7311-82	1 µl per sample	FACS
Anti-PD-1	BioXcell, West Lebanon, NH, USA	BE0273	200 µg/mouse/time	<i>in vivo</i> blockade

Supplementary Table S1.

Antibodies Used

Gene	Species	Accession number	Forward (5' to 3')	Reverse (5' to 3')	Usage
<i>D8S1786</i>	human		CGAAAGATTGAGACCCCAT	GTTTCCACACCGAAGCC	Microsatellite PCR
<i>D8S1752</i>	human		TCCTGGATCAGGCAGAAA	TCAGAGTTGGGTGAGCGA	Microsatellite PCR
<i>LINE-1</i>	human	M80343.1	AAAGCCGCTCAACTACATGG	TGCTTTGAATGCGTCCCAGAG	RT-PCR
<i>PDLIM2</i>	human	NG_030435.1	CAAAGCCAAGGACGCTGACC	GCTCTGGCGGATCTTGCTCT	RT-PCR
<i>pdlim2</i>	mouse	Mutant primer	GCTACCCGTGATATTGCTGAAGAG		Genotyping PCR
		WT primer	GTCTTCTCTTCCTCAGGCATGGCG		
		Common primer	GCCAACATTGCCCCAGAAGACTCC		
<i>Amp-R</i>	Plasmid		AACTTTATCCGCCTCCAT	GAAGCCATACCAAACGAC	Plasmid detection

Supplementary Table S2.

Primers Used

	<i>PDLIM2</i> gene	
	Relative RNA level	Copy number
NL-20	1.000	2.000
CALU-6	0.010	1.021
H727	0.020	1.590
H460	0.001	1.341
H2122	0.042	2.038
H23	0.128	0.376
H358	7.260	1.244
SKLU-1	0.390	1.685
SW1573	0.010	0.842
CALU-1	1.950	1.293
128-88T	0.049	1.767
H1299	0.003	0.243
273T	0.815	2.172
H827	1.017	1.237
H1650	2.168	0.679
H3255	5.118	0.587
343T	1.365	0.919
CALU-3	0.377	1.290
H1435	1.198	0.915
H1793	6.833	1.637
H596	0.195	0.782
H838	0.080	1.740
H1838	2.360	1.603
A549	0.010	5.265
H1975	1.086	2.507

Supplementary Table S3.

PDLIM2 RNA level and copy number in human lung cancer cell lines

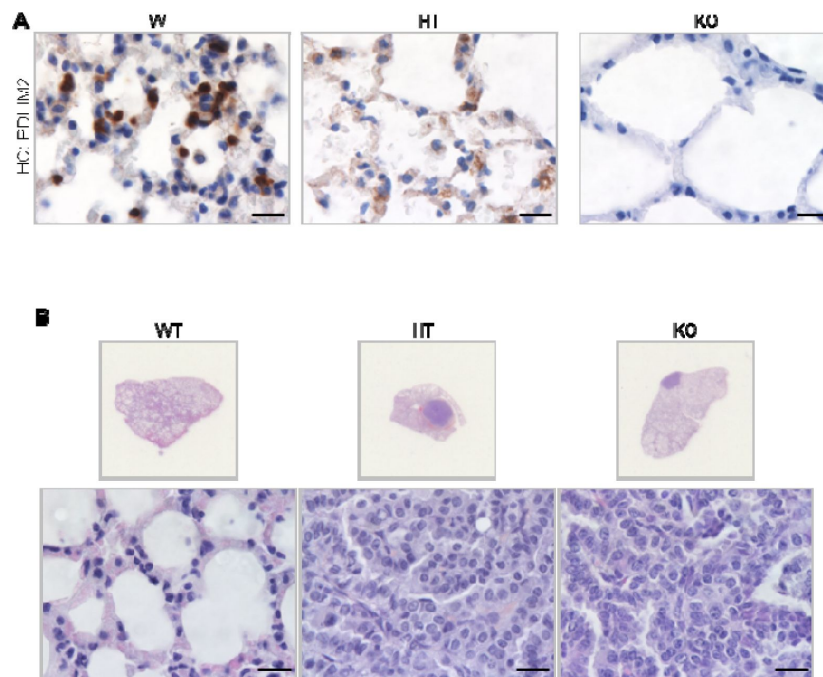


Figure S1.

Mice with *Pdlim2* deletion develop spontaneous tumors, majorly lung tumors. A, IHC assays showing decreased and complete loss of PDLIM2 protein expression in the lungs of *Pdlim2*^{+/-} (HT) and *Pdlim2*^{-/-} (KO) mice, respectively. Scale bar, 20 μ m. B, H&E staining of the lung tissues showing spontaneous lung tumors in *Pdlim2*^{+/-} and *Pdlim2*^{-/-} mice. Scale bar, 20 μ m.

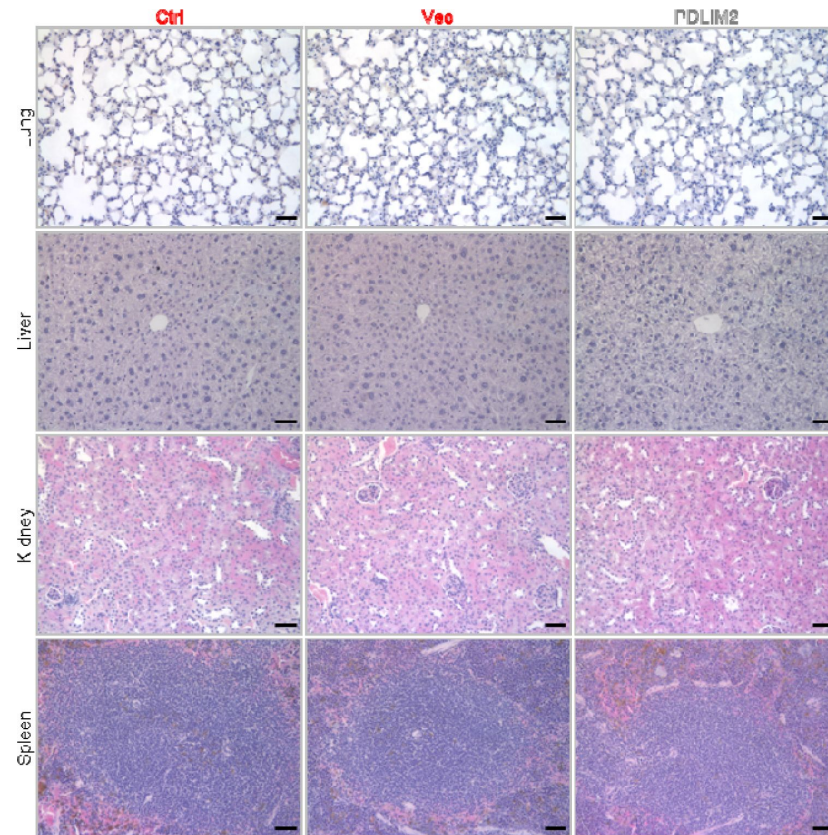


Figure S2.

PDLIM2 nanotherapy causes no obvious pathological changes in major organs. H&E staining showing comparable patterns in lung, liver, kidney and spleen between Vec and PDLIM2 group in the context of combinational chemotherapy and PD-1 blockade immunotherapy (n = 5). Scale bar: 50 μ m.

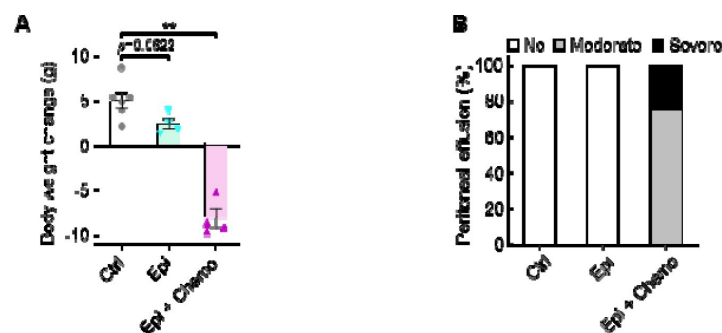


Figure S3.

Epigenetic drugs cause body weight loss in mice with lung cancer. Mice were i.p. injected with urethane (1 g/kg) for 6 weeks to induce lung tumors, and then treated for 6 weeks with 5-aza-dC and MS-275 (Epi, i.p., 1 mg/kg each, twice per week), carboplatin and paclitaxel (Chemo, i.p., 30 mg/kg and 15 mg/kg, respectively, once per week), or their combinations. A, Body weight change between prior to the first treatment and the euthanasia endpoint. B, Percentage of mice with peritoneal effusion (n $\geq$ 4).

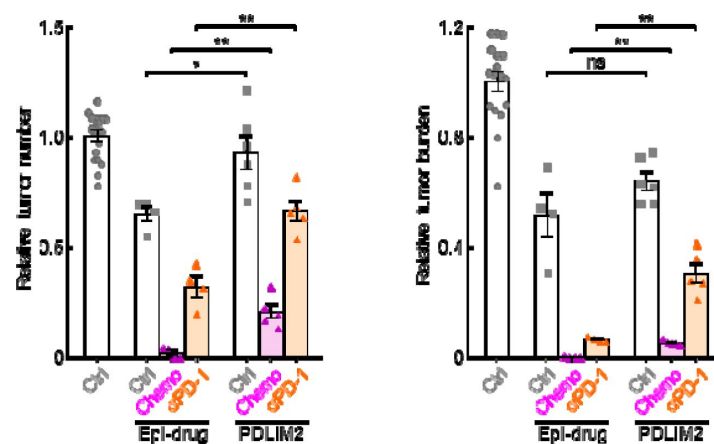


Figure S4.

Epigenetic drugs show better efficacy than PDLIM2 nanotherapy in lung cancer treatment. Mice were i.p. injected with urethane (1 g/kg) for 6 weeks to induce lung tumors, and then treated with 5-aza-dC and MS-275 (Epi, i.p., 1 mg/kg each, twice per week), carboplatin and paclitaxel (Chemo, i.p., 30 mg/kg and 15 mg/kg, respectively, once per week), anti-PD-1 antibody (aPD-1, i.p., 200 µg/mouse, three times per week), PDLIM2-expression plasmids containing nanoparticles (i.v., 25 µg plasmid/mouse, once per week), or their combinations for 6 weeks before they were sacrificed for tumor examination (n ≥ 4).

References

- [1] Siegel RL, Miller KD, Fuchs HE, Jemal A (2022) **Cancer statistics, 2022** *CA Cancer J Clin* **72**:7–33
- [2] Doroshow DB, Sanmamed MF, Hastings K, Politi K, Rimm DL, Chen L, Melero I, Schalper KA, Herbst RS (2019) **Immunotherapy in Non-Small Cell Lung Cancer: Facts and Hopes** *Clin Cancer Res* **25**:4592–4602
- [3] Zou W, Wolchok JD, Chen L (2016) **PD-L1 (B7-H1) and PD-1 pathway blockade for cancer therapy: Mechanisms, response biomarkers, and combinations** *Sci Transl Med* **8**
- [4] Zappasodi R, Merghoub T, Wolchok JD (2018) **Emerging Concepts for Immune Checkpoint Blockade-Based Combination Therapies** *Cancer Cell* **34**
- [5] Baxevasanos P, Mountzios G (2018) **Novel chemotherapy regimens for advanced lung cancer: have we reached a plateau?** *Ann Transl Med* **6**
- [6] Sun F, Li L, Yan P, Zhou J, Shapiro SD, Xiao G, Qu Z (2019) **Causative role of PDLIM2 epigenetic repression in lung cancer and therapeutic resistance** *Nat Commun* **10**
- [7] Garassino MC *et al.* (2020) **Patient-reported outcomes following pembrolizumab or placebo plus pemetrexed and platinum in patients with previously untreated, metastatic, non-squamous non-small-cell lung cancer (KEYNOTE-189): a multicentre, double-blind, randomised, placebo-controlled, phase 3 trial** *Lancet Oncol* **21**:387–397
- [8] Leonetti A, Wever B, Mazzaschi G, Assaraf YG, Rolfo C, Quaini F, Tiseo M, Giovannetti E (2019) **Molecular basis and rationale for combining immune checkpoint inhibitors with chemotherapy in non-small cell lung cancer** *Drug Resist Updat* **46**
- [9] Sun F, Guo ZS, Gregory AD, Shapiro SD, Xiao G, Qu Z (2020) **Dual but not single PD-1 or TIM-3 blockade enhances oncolytic virotherapy in refractory lung cancer** *J Immunother Cancer* **8**
- [10] Guo ZS, Qu Z (2021) **PDLIM2: Signaling pathways and functions in cancer suppression and host immunity** *Biochim Biophys Acta Rev Cancer* **1876**
- [11] Qu Z, Fu J, Yan P, Hu J, Cheng SY, Xiao G (2010) **Epigenetic repression of PDZ-LIM domain-containing protein 2: implications for the biology and treatment of breast cancer** *J Biol Chem* **285**:11786–11792
- [12] Qu Z, Yan P, Fu J, Jiang J, Grusby MJ, Smithgall TE, Xiao G (2010) **DNA methylation-dependent repression of PDZ-LIM domain-containing protein 2 in colon cancer and its role as a potential therapeutic target** *Cancer Res* **70**:1766–1772
- [13] Sun F, Xiao Y, Qu Z (2015) **Oncovirus Kaposi sarcoma herpesvirus (KSHV) represses tumor suppressor PDLIM2 to persistently activate nuclear factor kappaB (NF-kappaB) and STAT3 transcription factors for tumorigenesis and tumor maintenance** *J Biol Chem* **290**:7362–7368
- [14] Yan P, Qu Z, Ishikawa C, Mori N, Xiao G (2009) **Human T-cell leukemia virus type I-mediated repression of PDZ-LIM domain-containing protein 2 involves DNA methylation but independent of the viral oncoprotein tax** *Neoplasia* **11**:1036–1041

- [15] Vanoirbeek E, Eelen G, Verlinden L, Carmeliet G, Mathieu C, Bouillon R, O'Connor R, Xiao G, Verstuyf A (2014) **PDLIM2 expression is driven by vitamin D and is involved in the pro-adhesion, and anti-migration and -invasion activity of vitamin D** *Oncogene* **33**:1904–1911
- [16] Yan P, Fu J, Qu Z, Li S, Tanaka T, Grusby MJ, Xiao G (2009) **PDLIM2 suppresses human T-cell leukemia virus type I Tax-mediated tumorigenesis by targeting Tax into the nuclear matrix for proteasomal degradation** *Blood* **113**:4370–4380
- [17] Fu J, Yan P, Li S, Qu Z, Xiao G (2010) **Molecular determinants of PDLIM2 in suppressing HTLV-I Tax-mediated tumorigenesis** *Oncogene* **29**:6499–6507
- [18] Torrado M, Senatorov VV, Trivedi R, Fariss RN, Tomarev SI (2004) **Pdlim2, a novel PDZ-LIM domain protein, interacts with alpha-actinins and filamin A** *Invest Ophthalmol Vis Sci* **45**:3955–3963
- [19] Tanaka T, Soriano MA, Grusby MJ (2005) **SLIM is a nuclear ubiquitin E3 ligase that negatively regulates STAT signaling** *Immunity* **22**:729–736
- [20] Loughran G, Healy NC, Kiely PA, Huigsloot M, Kedersha NL, O'Connor R (2005) **Mystique is a new insulin-like growth factor-I-regulated PDZ-LIM domain protein that promotes cell attachment and migration and suppresses Anchorage-independent growth** *Mol Biol Cell* **16**:1811–1822
- [21] Tanaka T, Grusby MJ, Kaisho T (2007) **PDLIM2-mediated termination of transcription factor NF-kappaB activation by intranuclear sequestration and degradation of the p65 subunit** *Nat Immunol* **8**:584–591
- [22] Qu Z, Fu J, Ma H, Zhou J, Jin M, Mapara MY, Grusby MJ, Xiao G (2012) **PDLIM2 restricts Th1 and Th17 differentiation and prevents autoimmune disease** *Cell Biosci* **2**
- [23] Zhou J, Qu Z, Yan S, Sun F, Whitsett JA, Shapiro SD, Xiao G (2015) **Differential roles of STAT3 in the initiation and growth of lung cancer** *Oncogene* **34**:3804–3814
- [24] Qu Z, Sun F, Zhou J, Li L, Shapiro SD, Xiao G (2015) **Interleukin-6 Prevents the Initiation but Enhances the Progression of Lung Cancer** *Cancer Res* **75**:3209–3215
- [25] Steinbrecher KA, Harmel-Laws E, Sitcheran R, Baldwin AS (2008) **Loss of epithelial RelA results in deregulated intestinal proliferative/apoptotic homeostasis and susceptibility to inflammation** *J Immunol* **180**:2588–2599
- [26] Yu H, Pardoll D, Jove R (2009) **STATs in cancer inflammation and immunity: a leading role for STAT3** *Nat Rev Cancer* **9**:798–809
- [27] Xiao G, Fu J (2011) **NF-kappaB and cancer: a paradigm of Yin-Yang** *Am J Cancer Res* **1**:192–221
- [28] Li L, Sun F, Han L, Liu X, Xiao Y, Gregory AD, Shapiro SD, Xiao G, Qu Z (2021) **PDLIM2 repression by ROS in alveolar macrophages promotes lung tumorigenesis** *JCI Insight* **6**
- [29] Sun F, Li L, Xiao Y, Gregory AD, Shapiro SD, Xiao G, Qu Z (2021) **Alveolar Macrophages Inherently Express Programmed Death-1 Ligand 1 for Optimal Protective Immunity and Tolerance** *J Immunol* **207**:110–114

- [30] Wistuba II, Behrens C, Virmani AK, Milchgrub S, Syed S, Lam S, Mackay B, Minna JD, Gazdar AF (1999) **Allelic losses at chromosome 8p21-23 are early and frequent events in the pathogenesis of lung cancer** *Cancer Res* **59**:1973–1979
- [31] Kang J (2015) **Genomic alterations on 8p21-p23 are the most frequent genetic events in stage I squamous cell carcinoma of the lung** *Exp Ther Med* **9**:345–350
- [32] Macartney-Coxson DP *et al.* (2008) **Metastatic susceptibility locus, an 8p hot-spot for tumour progression disrupted in colorectal liver metastases: 13 candidate genes examined at the DNA, mRNA and protein level** *BMC Cancer* **8**
- [33] Swalwell JI *et al.* (2002) **Determination of a minimal deletion interval on chromosome band 8p21 in sporadic prostate cancer** *Genes Chromosomes Cancer* **33**:201–205
- [34] Wurster KD *et al.* (2017) **Inactivation of the putative ubiquitin-E3 ligase PDLIM2 in classical Hodgkin and anaplastic large cell lymphoma** *Leukemia* **31**:602–613
- 35. Matouk I, Raveh E, Ohana P, Lail RA, Gershtain E, Gilon M, De Groot N, Czerniak A, Hochberg A (2013) **The increasing complexity of the oncofetal h19 gene locus: functional dissection and therapeutic intervention** *Int J Mol Sci* **14**:4298–4316
- [36] Buscail L *et al.* (2015) **First-in-man phase 1 clinical trial of gene therapy for advanced pancreatic cancer: safety, biodistribution, and preliminary clinical findings** *Mol Ther* **23**:779–789
- [37] Nyamay'Antu A, Dumont M, Kedinger V, Erbacher p (2019) **Non-Viral Vector Mediated Gene Delivery: the Outsider to Watch Out For in Gene Therapy** *Cell & Gene Therapy Insights* **5**
- [38] Bonnet ME, Erbacher P, Bolcato-Bellemin AL (2008) **Systemic delivery of DNA or siRNA mediated by linear polyethylenimine (L-PEI) does not induce an inflammatory response** *Pharm Res* **25**:2972–2982
- [39] Yang J, Hendricks W, Liu G, McCaffery JM, Kinzler KW, Huso DL, Vogelstein B, Zhou S (2013) **A nanoparticle formulation that selectively transfects metastatic tumors in mice** *Proc Natl Acad Sci U S A* **110**:14717–14722
- [40] Kellar A, Egan C, Morris D (2015) **Preclinical Murine Models for Lung Cancer: Clinical Trial Applications** *Biomed Res Int* **2015**
- [41] Zhou J, Qu Z, Sun F, Han L, Li L, Yan S, Stabile LP, Chen LF, Siegfried JM, Xiao G (2017) **Myeloid STAT3 Promotes Lung Tumorigenesis by Transforming Tumor Immunosurveillance into Tumor-Promoting Inflammation** *Cancer Immunol Res* **5**:257–268
- [42] Li L, Han L, Sun F, Zhou J, Ohaegbulam KC, Tang X, Zang X, Steinbrecher KA, Qu Z, Xiao G (2018) **NF-kappaB RelA renders tumor-associated macrophages resistant to and capable of directly suppressing CD8(+) T cells for tumor promotion** *Oncimmunology* **7**
- [43] Sun F, Qu Z, Xiao Y, Zhou J, Burns TF, Stabile LP, Siegfried JM, Xiao G (2016) **NF-kappaB1 p105 suppresses lung tumorigenesis through the Tpl2 kinase but independently of its NF-kappaB function** *Oncogene* **35**:2299–2310
- [44] Hecht SS (2002) **Cigarette smoking and lung cancer: chemical mechanisms and approaches to prevention** *Lancet Oncol* **3**:461–469

- [45] Commisso C *et al.* (2013) **Macropinocytosis of protein is an amino acid supply route in Ras-transformed cells** *Nature* **497**:633–637
- [46] Huang JL *et al.* (2017) **Lipoprotein-biomimetic nanostructure enables efficient targeting delivery of siRNA to Ras-activated glioblastoma cells via macropinocytosis** *Nat Commun* **8**
- [47] Shi J, Kantoff PW, Wooster R, Farokhzad OC (2017) **Cancer nanomedicine: progress, challenges and opportunities** *Nat Rev Cancer* **17**:20–37

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Reviewer #1 (Public Review):

The manuscript by Sun and colleagues followed on their previous findings on the tumor suppressive role of PDLIM2 in lung cancer. They further investigated various mechanisms, including epigenetic modification, copy number variation and LOH, that led to the decrease expression of PDLIM2 in human lung cancer. Next, they used nanoparticle-based approach to specifically restore the expression in mouse lung tumors. They showed that over-expression PDLIM2 in lung cancer repressed its progression in vivo. Also, this treatment could synergize with chemotherapy and checkpoint inhibitor anti-PD-1. Overall, the results were quite promising and convincing, using a treatment combination that would appear to have potential for clinical implementation.

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Reviewer #2 (Public Review):

Summary: The authors have previously demonstrated that the E3 ligase PDLIM2 inhibits NF- κ B and STAT3 and is epigenetically repressed in human lung cancers (Sun et al. Nat. Comm. 2019 10: 5324); therefore, PDLIM2 is a tumor suppressor in lung cancer. In this manuscript, they follow up on their previous findings and show that expression of PDLIM2 is downregulated in human lung cancers by both genetic deletion and promoter methylation. They further describe a novel approach to restore the expression of PDLIM2 in mouse lung tumors by systemically administering PDLIM2 plasmids encapsulated in nanoparticles (termed "nanoPDLIM2"). The nanoPDLIM2 approach was shown to exhibit efficacy with low toxicity in a urethane-induced mouse lung cancer model. The authors further demonstrated synergy of nanoPDLIM2 with chemotherapy and PD-1 blockade immunotherapy. The combination therapy of nanoPDLIM2, chemotherapy and immunotherapy proved most

effective with complete tumor remission in 60% of mice. Mechanistically, nanoPDLIM2 upregulated MHC-I expression, enhanced CD4/CD8 T cell activation and tumor infiltration, and suppressed MDR1 induction and nuclear expression of STAT3, RelA and pro-survival genes in tumors. Overall, this study is important because it reinforces the critical roles of PDLIM2 in suppressing lung cancer, and also identifies a potential approach to restoring PDLIM2 expression in lung tumors. The experiments were well executed; the data are convincing and support the conclusions made by the authors.

<https://doi.org/10.7554/eLife.89638.2.sa0>

Author Response

Reviewer #3 (Public Review):

Strengths:

NanoPDLIM2, nanotechnologies that efficiently deliver lentivirus overcomes resistance to chemotherapy and anti-PD-1 immunotherapy. This is a new strategy for enhancing the efficiency of immune checkpoint inhibitors.

This finding is important from a clinical translation perspective, but I have several minor concerns.

Weaknesses:

1. Please describe the mechanism of increased MHC class I and PD-L1 by PDLIM2.

Our previous studies showed that PDLIM2 induces MHC-I induction through decreasing STAT3 whereas it is dispensable for PD-L1 expression (Sun et al, 2019, PMID: 31757943). In line with the studies, PD-L1 is induced by chemotherapeutic drugs, but not by NanoPDLIM2 (Figure 6A). Together with the roles of PDLIM2 in repressing RelA-dependent MDR1 induction by chemotherapy and in preventing expression of cell survival and proliferation genes by targeting both RelA and STAT3 (Sun et al, 2019, PMID: 31757943), further providing the mechanistic basis for the combination and synergistic effect of nanoPDLIM2, anti-PD-1 and chemo drugs. The improvement has now been further incorporated.

1. Please describe the mechanism of decreased MDR1, nuclear RelA and STAT3 by PDLIM2.

Our previous studies demonstrated that PDLIM2 reduces MDR1 expression by degrading nuclear RelA (Sun et al, 2019, PMID: 31757943).

1. Please determine whether PDLIM2 expression directly impacts immune cells (function and number)?

As shown in Figure 5, NanoPDLIM2 increased the number and activation of tumor infiltrating lymphocytes (TILs); and in prior study, PDLIM2 knockout repressed the numbers of TILs and inhibited the activation of CD4+ and CD8+ T cells, while its re-expression in lung tumors led to T cell activation (Sun et al. 2019, PMID: 31757943). On the other hand, selective deletion of PDLIM2 in immune cells and in particular myeloid cells repressed the numbers and activation of TILs (Li et al, 2021, PMID: 33539325; PMCID: PMC8021114). Thus, PDLIM2 may impact immune cells both directly and indirectly, particularly when nanoparticles can deliver PDLIM2 into both tumor cells and tumor-associated immune cells (despite PDLIM2 is delivered into much fewer immune cells compared to tumor cells).

1. *What is the efficiency of PDLIM2 delivery? Does delivery efficiency determine anti-tumor effect?*

As shown in the manuscript, the dose of PDLIM2 used already shows high delivery (20-30 copies per tumor cell in Figure 3B) and therapeutic efficacy in the mouse model of refractory lung cancer and particularly when being combined with anti-PD-1 and chemo drugs. It is of interest to test different doses in the model for the best delivery and efficacy, which is actively being pursued in the lab.

1. *Authors used a non-immunogenic tumor model. Can you demonstrate the combination effect with PDLIM2 in immunogenic lung cancer models to determine whether the combination of PDLIM2 with anti-PD-1 Ab confers a synergistic effect without chemotherapy?*

Yes, it is of interest to demonstrate the combination of PDLIM2 and anti-PD-1 in immunogenic lung cancer models with chemotherapy although a synergy is highly expected. The greatest challenge in the lung cancer field is the low response of non-immunogenic tumor, which is the focus of the current manuscript.

1. *On page 11, % change can make one over-interpret data.*

The % change has been removed from the manuscript.

1. *In Figure 5, what is the difference between 5A and 5D?*

Figure 5A shows the increase of TILs by nanoPDLIM2 in animals that did not receive PD-1 blockade immunotherapy, Figure 5D shows the increase of TILs by nanoPDLIM2 in animals received PD-1 blockade immunotherapy.

1. *It is unclear whether PDLIM2 confers an additive or a synergistic effect with anti-PD-1/chemo.*

PDLIM2 nanotherapy confers a synergistic effect with chemotherapy on increasing apoptosis in tumors (Figure 4B) and tumor reduction (Figure 4A and 6E, left panel, tumor number), confers a synergistic effect with antiPD-1 on increasing CD4⁺ and CD8⁺ TILs (Figure 5A and 5D), and apoptosis in tumors (Figure 5F), and an additive effect on tumor reduction (Figure 5C and 6E), and confers a synergistic effect with chemotherapy plus anti-PD-1 on increasing CD4⁺ and CD8⁺ TILs (Figure 5A and 6F) and tumor reduction (Figure 6E, left panel, tumor number).

1. *Have the authors tested any toxicity in normal lungs?*

Same to tumor lungs, no obvious toxicity has been observed in normal lungs.

Reviewer #1 (Recommendations For The Authors):

The paper is clear and well-written, although some minor edits are needed. For example, the title could be changed to reflect both human and mouse studies in the manuscript for more general readers. Moreover, 'lung cancer' should be used instead of 'lung cancers'. The manuscript could be further improved by validating their findings in a different model and particularly the syngeneic model of metastatic lung cancer for a

better overall survival time by the new combination therapy, given the fact that clinical trial studies usually start in patients with metastatic tumors. But this is optional because the therapeutic effect on primary lung cancer is already significant.

Thanks for the correction and wonderful suggestions. The “lung cancers” were replaced with “lung cancer”, and the title was changed to “Improving PD-1 blockade plus chemotherapy for complete remission of lung cancer by nanoPDLIM2”.

Reviewer #2 (Recommendations For The Authors):

1. *What is the rationale for i.v. injection of nanoparticles containing PDLIM2 plasmid? Intranasal administration of nanoparticles may potentially target nanoPDLIM2 specifically to the lungs. Another potential option is intranasal infection of mice with adenovirus expressing PDLIM2.*

The rationale for i.v. injection of nanoPDLIM2 is that iv injected nanoPDLIM2 first reach into the lung and more importantly tumor tissues as well as the convenience and high efficacy of mouse i.v. injection, particularly when multiple injections are needed. Mice are much less stressful compared to other intranasal or even intratracheal injection. Adenovirus can be used only once, because it will initiate ant-viral immune response in mice.

1. *The authors examine PDLIM2 expression in lung tumors 1 week after i.v. administration of nanoparticles (Fig. 3A). Do all tumor cells express PDLIM2 after nanoPDLIM2 treatment? How long does PDLIM2 persist in the tumors? The kinetics of PDLIM2 expression may be informative to help interpret the results from the various combination treatments given to the mice. Multiple rounds of nanoPDLIM2 treatment could potentially improve the efficacy of the treatment.*

For all the sections examined (n=6), PDLIM2 was re-expressed in most but not all lung cancer cells at 1-week of the i.v administration. Accordingly, nanoPDLIM2 was injected weekly. We are examining if PDLIM2 reexpression can last longer. We are also testing the best dose with the best efficacy.

1. *Does the plasmid DNA from nanoparticles trigger an innate immune response in the lung that contributes to anti-tumor responses?*

In line with previous studies showing no effect on immune responses (Bonnet et al. 2008. PMID: 18709489), the dose used in current study does not significantly affect immune cells in the lung, suggesting no obvious effect of nanoparticles with empty plasmid on innate immune response.

1. *In Fig. 4, does the combination of nanoPDLIM2 and chemotherapy diminish STAT3 nuclear staining?*

NanoPDLIM2 alone decreased nuclear STAT 3 in tumor cells (Figure 2C), it also diminished nuclear STAT3 in tumor cells with the combination of chemotherapy.